

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and implemented **Python**-based machine learning models for user targeting and personalization, achieving a **15% increase** in click-through rates for growth initiatives.
- Designed and built scalable **ML pipelines** using **Python** and **SQL** for data processing, model training, and deployment on **AWS**.
- Collaborated with cross-functional teams including product managers and data analysts to identify and scope **ML solutions** for user engagement challenges.
- Conducted **A/B tests** to evaluate the performance of personalization models, analyzing results to optimize impact on key growth metrics like retention.
- Analyzed large datasets using **SQL** and **Spark** to extract actionable insights informing user acquisition and retention strategies.
- Contributed to the development and maintenance of the team's **ML infrastructure**, focusing on efficient model deployment and monitoring on **AWS**.
- Containerized ML services using **Docker** and deployed them to an **AWS Kubernetes** cluster, managing configurations and resource allocation.

SKILLS

ML Concepts – Machine Learning, Recommendation Systems, User Modeling, Personalization Algorithms, Statistical Analysis, Model Evaluation

Programming Languages – Python, SQL

ML Frameworks & Libraries – PyTorch, scikit-learn, pandas, NumPy

Data Processing & Big Data – Spark, Hadoop, Data Warehousing, ETL

Cloud & DevOps – AWS, Docker, Kubernetes

Collaboration & Methodologies – Agile, A/B Testing, Code Reviews, Cross-functional Teams

PROJECTS

Personalized Recommendation Engine for User Engagement

- Developed a **Python**-based recommendation engine using **scikit-learn** collaborative filtering techniques to suggest relevant content to users.
- Engineered feature extraction pipelines using **SQL** and **pandas** from user interaction logs, processing over 1 million records to create user profiles.
- Trained and evaluated the recommendation model on a diverse user dataset, achieving a **0.82 Mean Reciprocal Rank (MRR)** and improving user session duration by **18%**.

Scalable ML Pipeline for User Segmentation

- Designed and implemented a CI/CD-integrated **ML pipeline** using **Python** and **AWS Lambda** to automate user segmentation based on behavioral data.
- Integrated with **PostgreSQL** for data retrieval and storage, ensuring data integrity and efficient processing for model retraining.
- Containerized the pipeline components with **Docker** for consistent deployment across development and production environments on **AWS**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)